

- BC 15. If $f'(x)$ and $g'(x)$ exist and $f'(x) > g'(x)$ for all real x , then the graph of $y = f(x)$ and the graph of $y = g(x)$

1969

NC

- (A) intersect exactly once.
(B) intersect no more than once.
(C) do not intersect.
(D) could intersect more than once.
(E) have a common tangent at each point of intersection.

- BC 34. Which of the following is an equation of a curve that intersects at right angles every curve of the family $y = \frac{1}{x} + k$ (where k takes all real values)?

1969

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- (A) $y = -x$ (B) $y = -x^2$ (C) $y = -\frac{1}{3}x^3$ (D) $y = \frac{1}{3}x^3$ (E) $y = \ln x$

- BC 4. For what non-negative value of b is the line given by $y = -\frac{1}{3}x + b$ normal to the curve $y = x^3$?

1973

NC

- (A) 0 (B) 1 (C) $\frac{4}{3}$ (D) $\frac{10}{3}$ (E) $\frac{10\sqrt{3}}{3}$

- BC 23. $\lim_{h \rightarrow 0} \frac{1}{h} \ln\left(\frac{2+h}{2}\right)$ is

1973

NC



- (A) e^2 (B) 1 (C) $\frac{1}{2}$ (D) 0 (E) nonexistent

- BC 8. If f is a function such that $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = 0$, which of the following must be true?

1985

NC

- (A) The limit of $f(x)$ as x approaches 2 does not exist.
(B) f is not defined at $x = 2$.
(C) The derivative of f at $x = 2$ is 0.
(D) f is continuous at $x = 0$.
(E) $f(2) = 0$

- BC 4. $\frac{d}{dx}(xe^{\ln x^2}) =$

1997

NC

- (A) $1 + 2x$ (B) $x + x^2$ (C) $3x^2$ (D) x^3 (E) $x^2 + x^3$